

Name: _____

UNIT 30: MOLECULAR GENETICS

Topic 3: Mutations

Mistakes in the coding of genetic information often take place during DNA replication, transcription or translation, resulting in the production of a different sequence of bases than what was intended. These mistakes are referred to as mutations.

- Mutations that occur in _____ (body cells) are not passed on to offspring (e.g. _____).
- Mutations that occur in the _____ (reproductive cells) are passed on to offspring (e.g. _____).

Causes of Mutations

Mutation Type	Definition	Example
Point		
Frameshift		

Name: _____

Effects of Mutations

List and explain the three types of effects a mutation can have on cell functioning:

1)

2)

3)

PRACTICE

Identify the cause & effect of each of the following mutations...

a. Original strand: AUG UUU UUG CCU UAU CAU CGU

Mutated strand: AUG UUU UUG CCU UAC CAU CGU

b. Original strand: AUG UUU UUG CCU UAU CAU CGU

Mutated strand: AUG UUU UUG CCU UAA CAU CGU

c. Original strand: AUG UUU UUG CCU UAU CAU CGU

Mutated strand: AUG UUU UUG CCU AUC AUC GU

Name: _____

Mitochondrial DNA

➤ *What is mtDNA? How does the inheritance of mtDNA differ from that of DNA?*

Recombinant DNA

A molecule of DNA that includes genetic material from different sources is called recombinant DNA. It is produced through the process of genetic engineering. Using a diagram, outline this process in the space below, including the role of plasmids, restriction endonucleases, DNA ligase and methylase:



Name: _____

Gel Electrophoresis

What it is	How it works	What it's used for

Taq Polymerase & PCR

What it is	How it works	What it's used for